

Agnostic Equalization Mechanism (AEM)

Technical Specification Summary: v1.0 (Production)

Status: TRL 5 Validated — Hardware-Isolated Release

Release Date: January 9, 2026

IP Status: Patent Pending (#63/938,607)

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Executive Abstract

AEM v1.0 marks the transition from software-based simulation to hardware-enforced deterministic safety. By relocating the Structural Veto logic into the ARM TrustZone Secure World, we eliminate the "OS-compromise" vulnerability inherent in traditional AI guardrails. This document verifies a 54.4 μ s mean response time, enabling real-time, hardware-isolated governance for autonomous agents.

1. System Overview

The AEM provides a deterministic "Structural Veto" for autonomous agents. By utilizing ARM TrustZone, the mechanism ensures that safety-critical logic is physically and cryptographically isolated from the host Operating System (Rich Execution Environment).

2. Technical Specifications (Verified Jan 9, 2026)

Parameter	Specification
Target Hardware	Broadcom BCM2837 (ARMv8-A)
TEE Environment	OP-TEE 3.16.0 (Secure World)
Inter-World Latency (Avg)	54.4 μs
Peak Throughput Latency	10.7 μs
Inspection Method	Deep Packet Inspection (DPI) / Semantic Invariant Match
Decision Logic	Deterministic (Non-Probabilistic)

3. The "Warm Path" Architecture

Unlike standard software-based firewalls, AEM utilizes a **Shared-Memory Bridge**.

- The Orchestrator** (Normal World) maps a secure buffer.
- The Monitor** (Secure World) performs a "hardware-look" into that buffer via a Trusted Application (TA).
- The Veto** is issued via an unmaskable secure return code (0xf0100001), bypassing standard OS interrupt delays.

4. TRL 5 Validation Results

During the January 2026 validation window, the AEM was subjected to **567 hardware-in-the-loop pulses**.

- **Success Rate:** 100% detection of AUTH_VETO command strings.
- **False Positive Rate:** 0.00% across safe command sets.
- **Jitter Stability:** Measured variance < 45 μ s, ensuring predictable real-time response.

5. Glossary of Terms

- **TEE (Trusted Execution Environment):** A secure area of a main processor that guarantees code and data loaded inside are protected with respect to confidentiality and integrity.
- **DPI (Deep Packet Inspection):** A method of examining the data part of a packet as it passes an inspection point.
- **Structural Veto:** A safety intervention based on the physical or logical structure of a command, rather than a probabilistic guess.

Contact & Citations

For technical inquiries or full validation datasets, please contact **James Balousek** via **Zenith Structural Holdings LLC**.

Cite as: Balousek, J. (2026). *Agnostic Equalization Mechanism (AEM): Structural Veto Kernel*. US Patent App #63/938,607.